INT421—Digital Signal Processing

Discrete-Time Signals and Systems

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Topics of Last Lecture

- Signal Definition
  - A function of one or more independent variables which conveys information on the nature of a physical phenomenon.
- Types of Signals
  - Single dimensional signal and Multi-dimensional signals.
- Analog to Digital Conversion
- Nyquist–Shannon Sampling Theorem

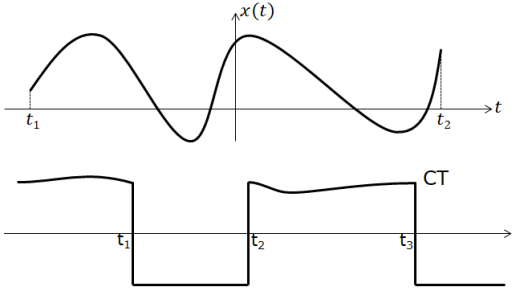


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1. Continuous-time vs. Discrete-time
2. Discrete-Time Signals
3. Signal Properties
4. Summary

Continuous-time vs. Discrete-time

- The independent variable may be either *continuous* or *discrete*.
- **Continuous-time signals**: defined at every instant of time over some time interval.
  - $x(t)$ where t can take any real value.
  - $x(t)$ may be 0 for a given range of values of t.
  - Values for x may be real or complex.
  - t_1, t_2, and t_3 are points of discontinuity.
  - CT is defined at a continuum points in time

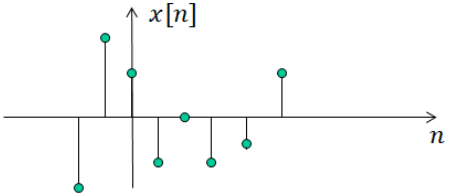


Continuous-time vs. Discrete-time

- **Discrete-time signals**: defined at discrete points in time and not between them.
- The independent variable takes only a discrete set of values $x[n]$ where

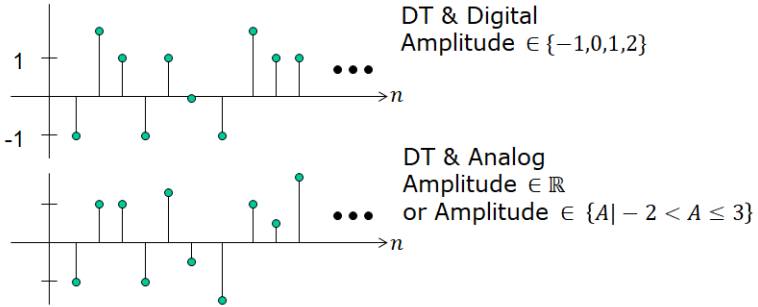
$$n \in \{\cdots -3, -2, -1, 0, 1, 2, 3 \ldots\}$$

- Values for x may be real or complex.



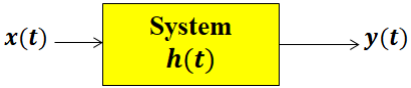
Continuous-time vs. Discrete-time

- **Finite set**: has a beginning and an end and the number of elements is countable.
  - Every finite set is countable
  - e.g., $\{-1, 0, 1, 2\}$, $\{x \in \mathbb{Z} : -2 < x \leq 2\}$
- **Infinite set**: does not has start or end or has both start and end but is uncountable.
  - e.g., infinite and countable: $\{\ldots, 2, 4, 6, 8\}$, $\{-1, 1, 3, 5, 7, \ldots\}$,$\{\ldots, 2, 5, 8, 11, 14, \ldots\}$.
- If the signal's amplitude (A) is a finite set $\Longrightarrow$ the signal is Digital, otherwise it is Analog.



Continuous-time vs. Discrete-time System

- Systems are classified according to the type of input signals and output signals.
- **Continuous-time system**: all the signals are CT signals.



- **Discrete-time system**: all the signals are DT signals.

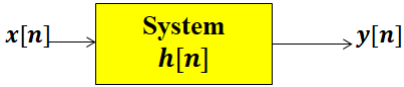


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1 Continuous-time vs. Discrete-time

2 Discrete-Time Signals

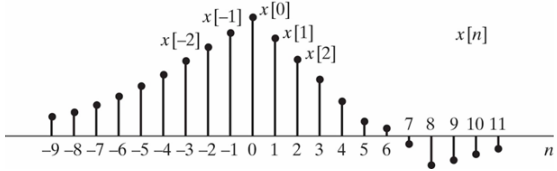
3 Signal Properties

4 Summary



Discrete-Time Signals

- Discrete-time signals are represented by sequence of numbers
  - The nth number in the sequence is represented with $x[n]$, also called the nth sample of the sequence.
  - The sequence $x = \{x[n]\} = \{\ldots, x[-2], x[-1], \underset{\uparrow}{x[0]}, x[1], x[2], \ldots\}, \quad n \in \mathbb{Z}$



  - $x[n]$ is used to refer to the whole sequence not just the nth sample.
  - Example: $x[n] = \{1.2, 0.09, \underset{\uparrow}{-2.23}, 5, 11, -2\}$

Using MATLAB

$n = [-2, -1, 0, 1, 2, 3]; \quad x = [1.2, 0.09, -2.23, 5, 11, -2]; \quad stem(n, x);$

Discrete-Time Signals

- There is a number of different ways to represent discrete-time signals:

Functional

$$x[n] = \begin{cases} \left(\frac{1}{2}\right)^n, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

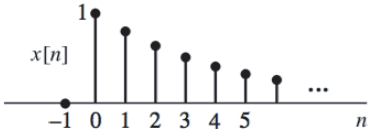
Tabular

n	$\ldots$	-2	-1	0	1	2	3	$\ldots$
$x[n]$	$\ldots$	0	0	1	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\ldots$

Sequence

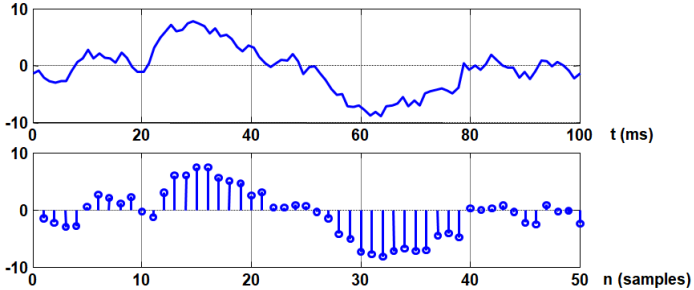
$$x[n] = \{ \ \ldots \ 0 \ \underset{\uparrow}{1} \ \tfrac{1}{2} \ \tfrac{1}{4} \ \tfrac{1}{8} \ \ldots \ \}$$

Pictorial



Discrete-Time Signals

- The times sequences are obtained by **sampling** of continuous-time signals.
  - In this case $x[n]$ is value of the analog signal at $x_c(nT)$
  - Where T is the sampling period.
  - $\frac{1}{T}$ is the sampling frequency.



Quiz

Examples

Sketch the signal $x[n] = 2^{-|n|}$ for $-5 < n < 5$

Quiz

Examples

Sketch the signal $x[n] = 2^{-|n|}$ for $-5 < n < 5$

Solution

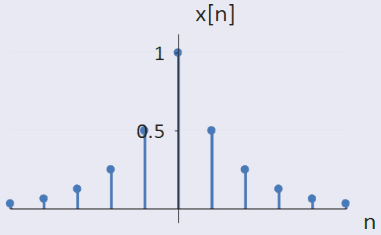


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1 Continuous-time vs. Discrete-time

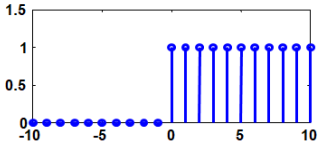
2 Discrete-Time Signals

3 Signal Properties

4 Summary

Deterministic vs. Random

- Deterministic signal are completely specified for any instant of time.
  - Can be described mathematically, or Graphically.
    - e.g. the Unit step $u[n]$
- Random signal amplitudes
  - Cannot be predicted exactly
  - Cannot be described by a mathematical function



Signal Symmetry: Even/Odd Signal

- Symmetric or Even signal: $x[n]$ is even if

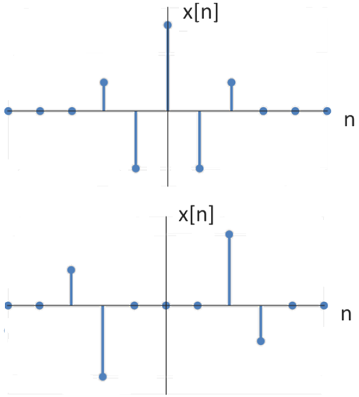
$$x[n] = x[-n]$$

$x[n]$ is symmetrical about the vertical axis "vertical axis acts as a mirror".

- Anti-symmetric or Odd signal

$$x[n] = -x[-n]$$

$x[n]$ is anti-symmetrical about the vertical axis.



Examples of Even/Odd Signal

- Even signal:

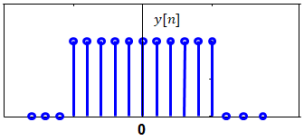
$$x[n] = \cos(\pi n) = \cos(-\pi n) = x[-n]$$

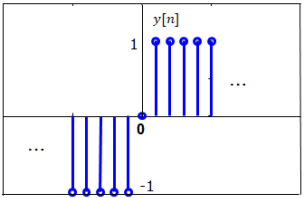
- Odd signal

$$x[n] = \sin(\pi n) = \sin(-\pi n) = -x[-n]$$

and

$$y[n] = sgn[n] = \begin{cases} 1, & n > 0 \\ 0, & n = 0 \\ -1, & n < 0 \end{cases}$$





Even/Odd Signal

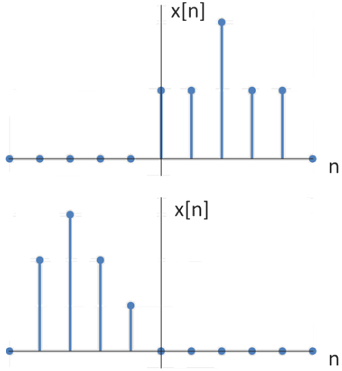
- Causal Signal

$$x[n] = 0, \quad n < 0$$

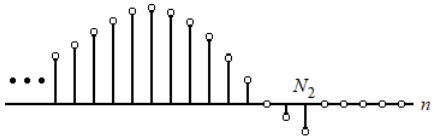
- Anti-Causal Signal

$$x[n] = 0, \quad n \geq 0$$

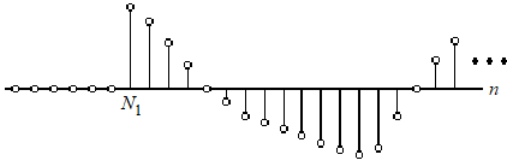
- Acausal signal
  - Has non-zero components before and after $n = 0$



Left- and Right-Sided



Left-sided

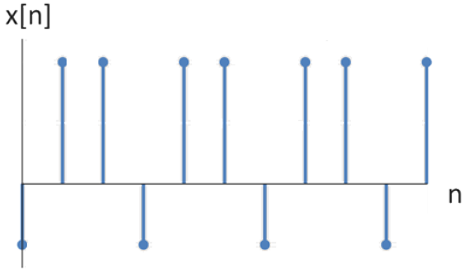


Right-sided

- $x[n]$ may be defined only for certain n:
  - $N_1 \leq n \leq N_2$: Finite length (length $=n$)
  - $N_1 \leq n$: Right-sided (**Causal** if $N_1 \geq 0$)
  - $n \leq N_2$: Left-sided (**Anticausal**)
- Can always extend with **zero-padding**

Periodic / Aperiodic

- Periodic Signal
  - $x[n] = x[n + \alpha N]$ where N and α are integers.
  - N is the fundamental period of the signal $x[n]$



Quiz

Examples

Consider the signal $x[n] = 2^{-|n|}$ for $-5 < n < 5$

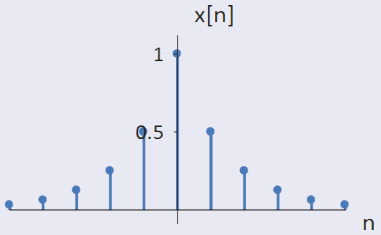
1. Sketch the signal.
2. Is the signal *even*, *odd*, or *neither*?
3. Is the signal *causal*, *anti-causal*, or *acausal*?
4. Is the signal periodic or *aperiodic*

Quiz

Solution

Consider the signal $x[n] = 2^{-|n|}$ *for* $-5 < n < 5$

1. *Sketch the signal.*

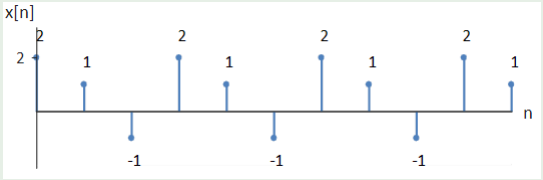


2. *Is the signal even, odd, or neither?* **Even**
3. *Is the signal causal, anti-causal, or acausal?* **Acausal**
4. *Is the signal periodic or aperiodic* **Aperiodic**.

Quiz

Examples

Consider the signal (assume pattern continues for $-\infty < n < \infty$)

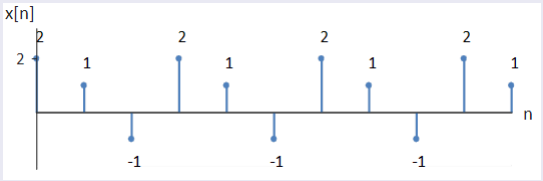


1. Is the signal even, odd, or neither?
2. Is the signal causal, anti-causal, or acausal?
3. Is the signal periodic or aperiodic?

Quiz

Solution

Consider the signal (assume pattern continues for $-\infty < n < \infty$)

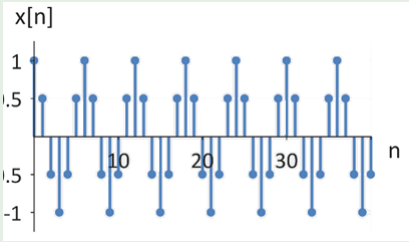


1. *Is the signal even, odd, or neither?* **Neither**
2. *Is the signal causal, anti-causal, or acausal?* **Acausal**
3. *Is the signal periodic or aperiodic?* **Periodic**

Quiz

Examples

$$x[n] = \cos((\pi/3)n)$$

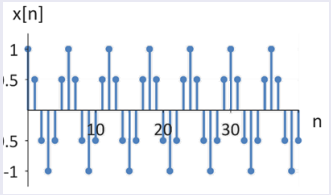


1. Is this signal periodic? if yes compute its periodicity and rational frequency.

Quiz

Solution

$$x[n] = \cos((\pi/3)n)$$



1. *Is this signal periodic?* **Yes**
   - *Period:* $N = 6$

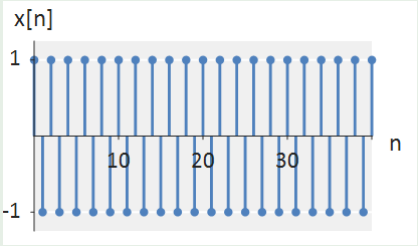
     $$\pi/3n + 2\pi = \pi/3(n+N) \Rightarrow 2\pi = \pi/3N$$

   - *Rational Frequency:* $f = 1/N = 1/6$

Quiz

Examples

$$x[n] = \cos(\pi n)$$

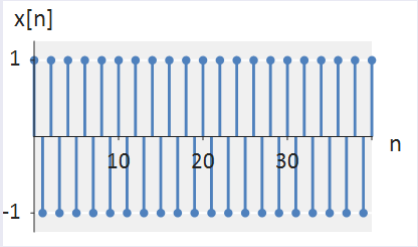


1. Is this signal periodic? if yes compute its periodicity and rational frequency.

Quiz

Solution

$$x[n] = \cos(\pi n)$$



1. *Is this signal periodic?* **Yes**
   - *Period:* $N = 2$
   - *Rational Frequency:* $f = 1/2$

Energy and Power of Signals

- Energy of a sequence is defined by

$$E = \sum_{-\infty}^{\infty} |x[n]|^2$$

- Power of a sequence $x[n]$ is defined as the average energy per sample

$$P = \lim_{n \longrightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^{N} |x[n]|^2$$

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Summary

- Signal Representations
- Signal Properties
  - Deterministic / Random.
  - Odd / even.
  - Periodic / Aperiodic.



Any Questions?

